Embeddings

Text → "what is the cost of solar panels"

Input (what is the cost of solar panels"
16 20 11 5 4 10 6
T1 T2 T3 T4 T5 T6 T7



What → 16 → numerical representation
↳ semantic meaning

Embedding

cost → 5 ⇒ [-0.1, 0.2, 0.3, 0.6] ← semantic meaning ← meaning

Embedding

Embeddings → meaningful vector representation of a word or token

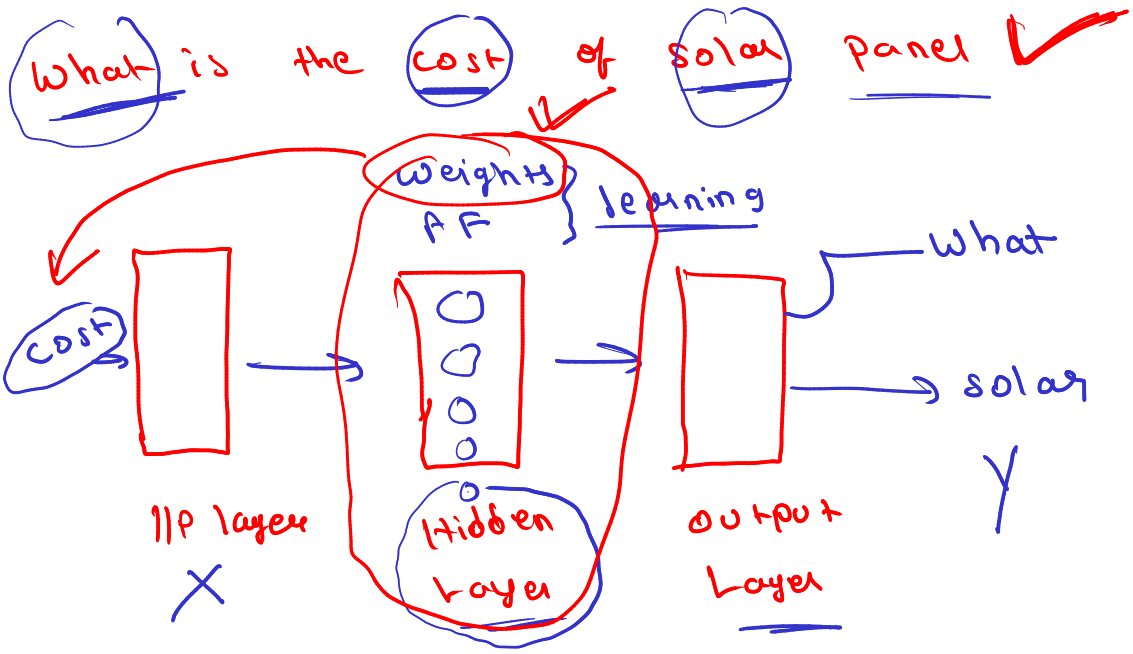
⇒ NN - neural Networks

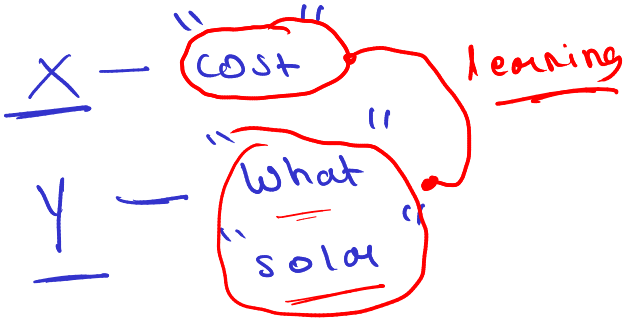
Embedding

⇒ 1. word2vec } word to vector
2. Glove

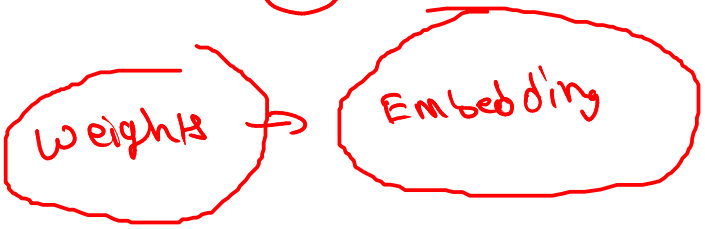
NN

⇒ skip gram
→ continous BOW

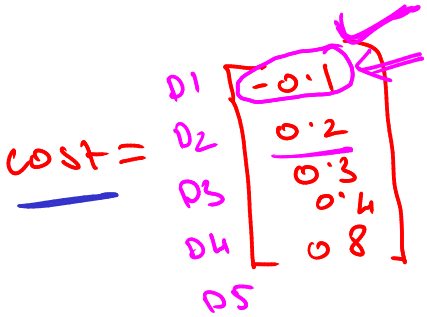




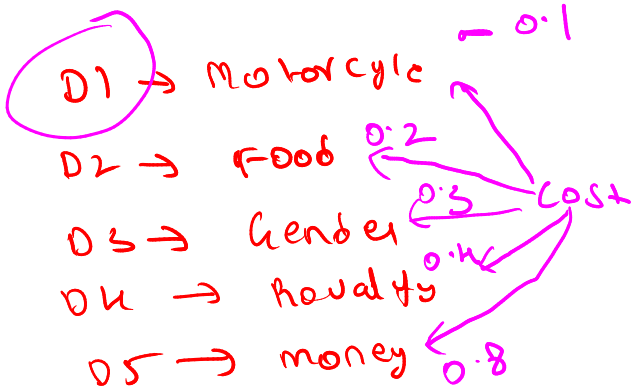
$$Y = W \cdot X + b$$

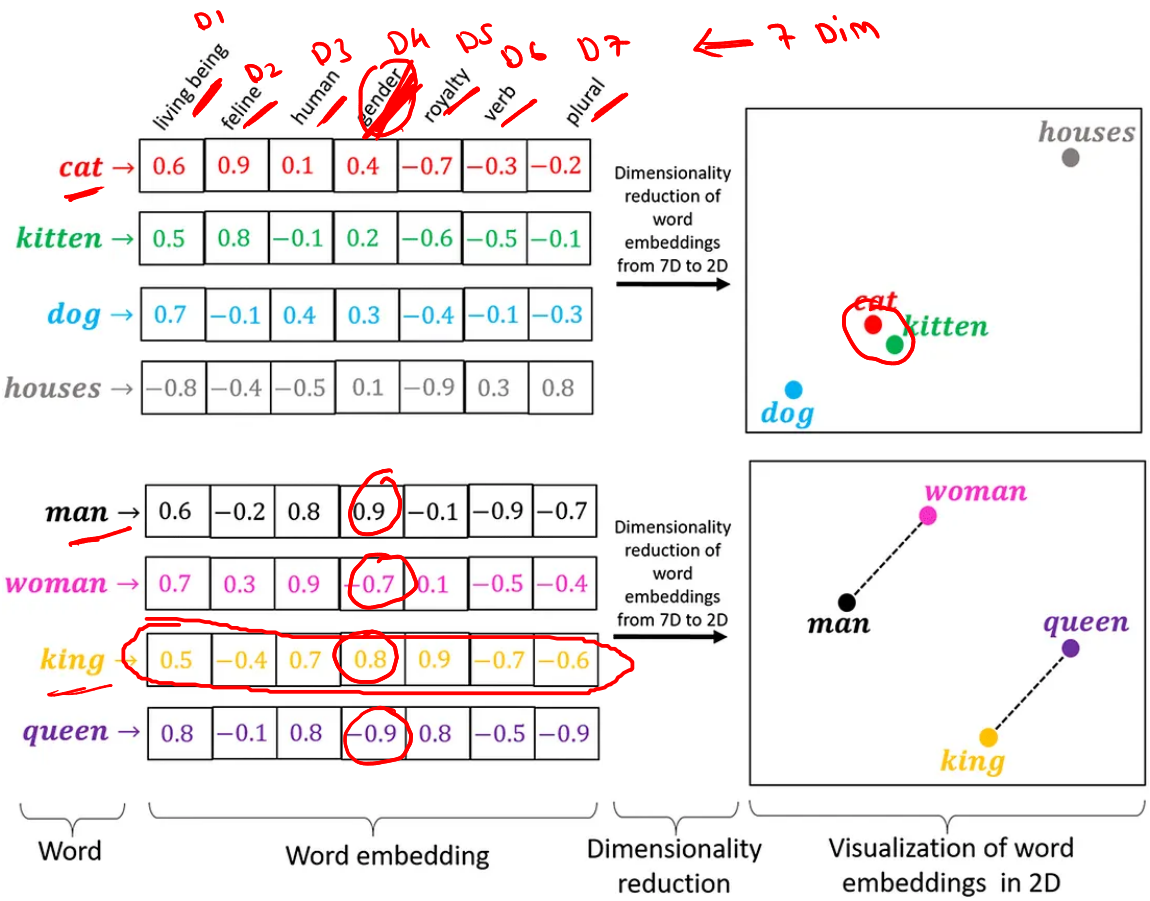


Embedding → Dimension ⇒ no of weights



Dimension





King

Queen

man

women

Word $\begin{bmatrix} D1 \\ D2 \\ \vdots \\ D7 \end{bmatrix}$

King − man + women ≈ queen

King ≈ queen

man ≈ women

Sentence2

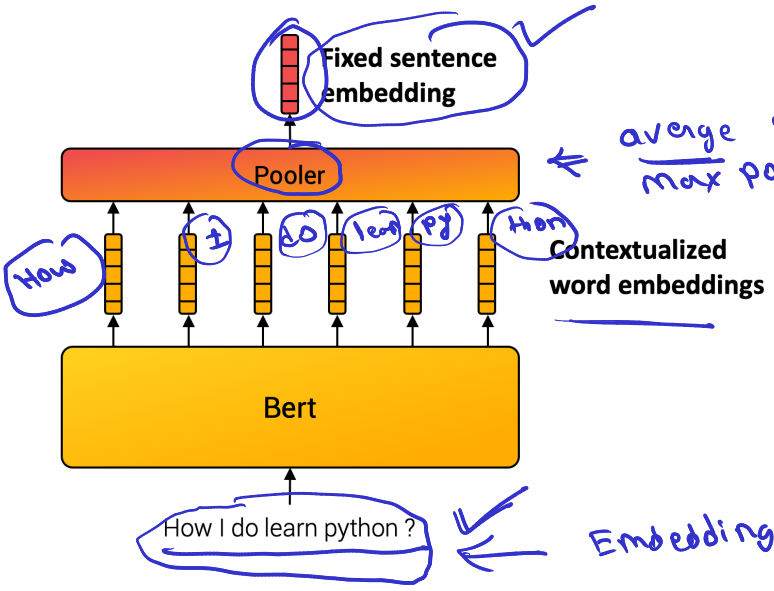
Sentence Embedding

1. BERT
2. SBERT
3. Transformer

BERT → Bidirection Encoder representation from Transform

Sentence transformer

"How do I learn python" → [ ]

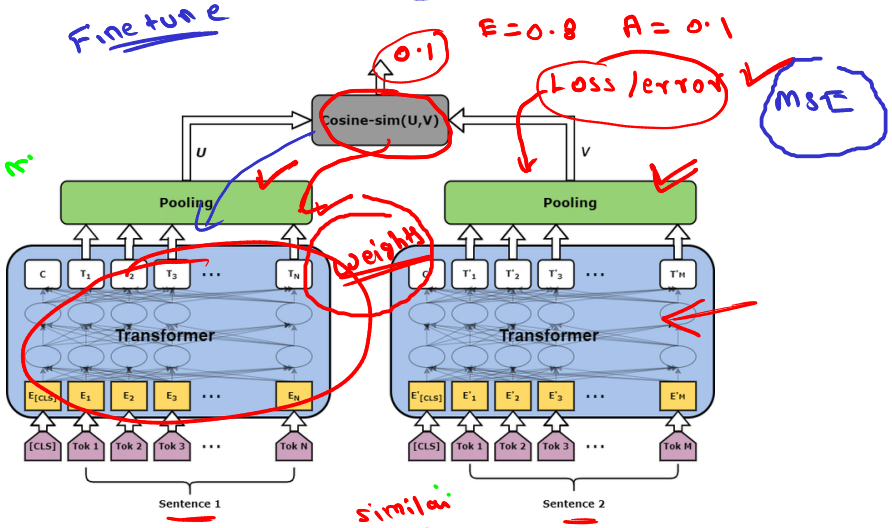


average pooling
max pooling

Embedding

Siamese Network

Finetune



E=0.8 A=0.1

Loss/error

MSE

weight

similar

⇒ How do I learn python? (S1) ⟷ Python is easy language (S2)

⇒ AI is the future (S1) ⟷ I love coffe (S2) ⇒ 0.2

non similar

RAG ⟶ Sentence Embedding ⇒ model

open source Embedding	closed source
1. Sentence transformer all-miniLM-v2 384 Dimension 2. gemma Embeddings 3. qwen3 - Embeddings	1. openAI Embedding ✓ 2. Gemini Embeddings 3. Anthropic

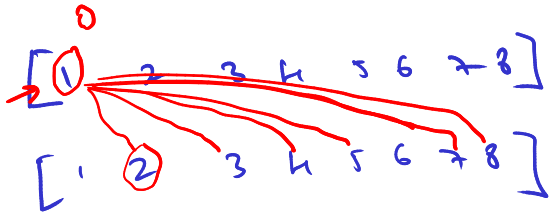
text-embedding-3-small — 1536 Dim

text-embedding-3-large — 3000 Dim

all-MiniLM-L6-v2 ⇒ Downloaded

↓

384 Dim

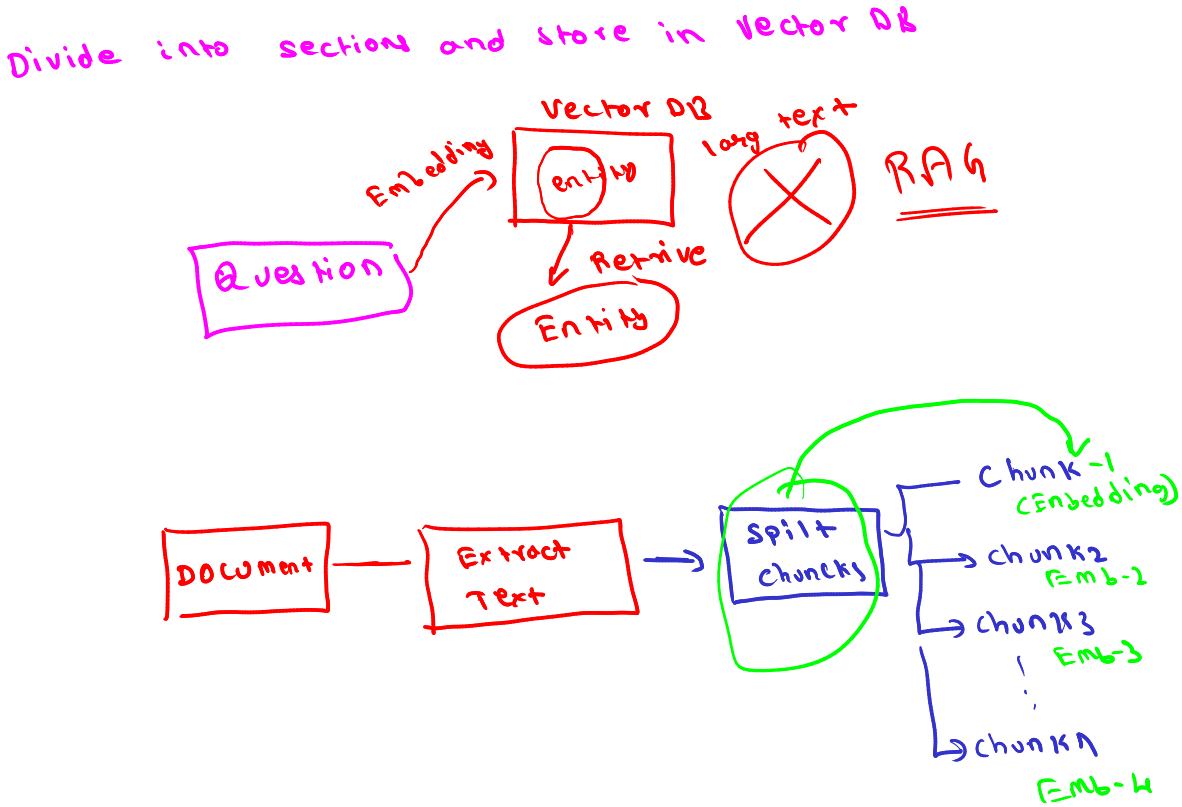


1 for

2nd for

```
texts = [
   "Machine learning algorithms",
   "Deep learning neural networks",
   "Natural language processing",
   "Computer vision applications",
   "Data science techniques",
   "AI is the future",
   "I love coffee",
   "Machine learning algorithms"
]
```

```
texts = [
   "Machine learning algorithms",
   "Deep learning neural networks",
   "Natural language processing",
   "Computer vision applications",
   "Data science techniques",
   "AI is the future",
   "I love coffee",
   "Machine learning algorithms"
]
```



Why Text Splitting Matters:

* LLM Context Limits: Most LLMs have token limits
* Semantic Preservation: Good splitting maintains meaning and context
* Retrieval Quality: Proper chunks improve vector search accuracy
* Processing Efficiency: Smaller chunks are easier to process and embed

CharacterTextSplitter -

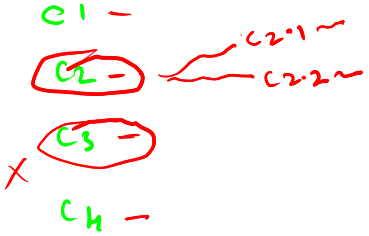
1. Split text using ONE separator
2. Iterate through the resulting splits
3. Greedily merge splits into chunks until chunk_size would be exceeded
4. If adding a split exceeds chunk_size:
   - finalize the current chunk
   - start a new chunk with the split
5. If a single split itself exceeds chunk_size:
   - keep it intact (chunk may exceed limit)
6. After all chunks are formed, apply chunk_overlap between consecutive chunks if possible

Use Cases:

1. Simple Documents: Plain text with clear paragraph breaks
2. Quick Prototyping: Fast setup for basic text processing
3. Uniform Content: Documents with consistent structure

RecursiveCharacterTextSplitter

1. Start with the first separator in the list
2. Split text at all occurrences of this separator
3. For each resulting piece:
   - If it's within chunk_size, keep it
   - If it's too large, try the next separator recursively
4. Continue until all pieces are within chunk_size or no separators remain
5. Add overlap between consecutive chunks



Artificial Intelligence (AI) has revolutionized the way we interact with technology.
Machine learning, a subset of AI, enables computers to learn from data without explicit programming.

Deep learning uses neural networks with multiple layers to process complex patterns.
Natural language processing (NLP) helps computers understand and generate human language.

Computer vision enables machines to interpret visual information from the world.
These technologies work together to create intelligent systems that can perform tasks previously thought to be exclusive to humans.

Robotics combines AI with mechanical engineering to create autonomous machines.
These robots can navigate environments, manipulate objects, and make decisions based on sensor data.

The future of AI holds promise for solving complex problems in healthcare, climate change, and scientific research. However, ethical considerations must be addressed to ensure responsible development and deployment of AI systems.

Use Cases:

1. Structured Documents: Markdown, technical docs, articles
2. Multi-level Content: Documents with headers, sections, subsections
3. Semantic Preservation: When maintaining context is crucial

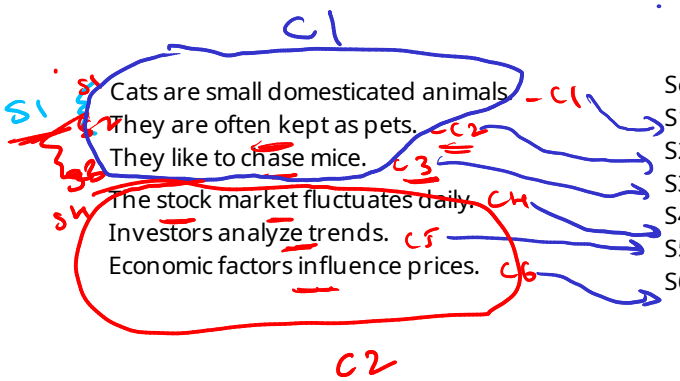
---

Semantic Chunker

- Cats are small domesticated animals. They are often kept as pets.
  They like to chase mice.

  The stock market fluctuates daily. Investors analyze trends.
  Economic factors influence prices.

by default "\n" — line by line

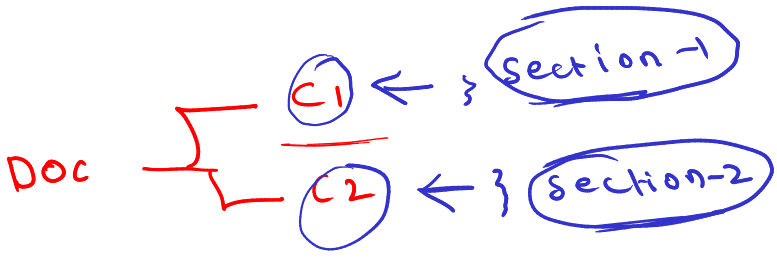


Sentence	Embedding
S1	[0.9, 0.8, 0.7]
S2	[0.88, 0.79, 0.69]
S3	[0.87, 0.78, 0.68]
S4	[0.2, 0.3, 0.4]
S5	[0.21, 0.31, 0.39]
S6	[0.22, 0.29, 0.38]

S1 → S2	0.99
S2 → S3	0.99
S3 → S4	0.45
S4 → S5	0.98
S5 → S6	0.97

Store the chunks

Embedding — Vector DB


C1
Section-1
Doc
C2
Section-2